2 The random variable X has probability generating function $G_X(t)$ given by

$$G_X(t) = \tfrac{1}{5} + pt + qt^2,$$

where p and q are constants.

(a) Given that $E(X) = 1.1$, find the numerical value of $\text{Var}(X)$. [4]

The random variable Y has probability generating function $G_Y(t)$ given by

$$G_Y(t) = \tfrac{2}{3}t\left(1 + \tfrac{1}{2}t^2\right).$$

The random variable Z is the sum of independent observations of X and Y.

(b) Find the probability generating function of Z. [2]

....

....

....

....

....

....

....

....

....

....

....

....

(c) Find $P(Z > 2)$. [1]

....

....

....

....

(d) State the most probable value of Z. [1]

....

....

....

....

....

....